

Beyond Brain Decoding: Visual-Semantic Reconstructions to Mental Creation Extension Based on fMRI

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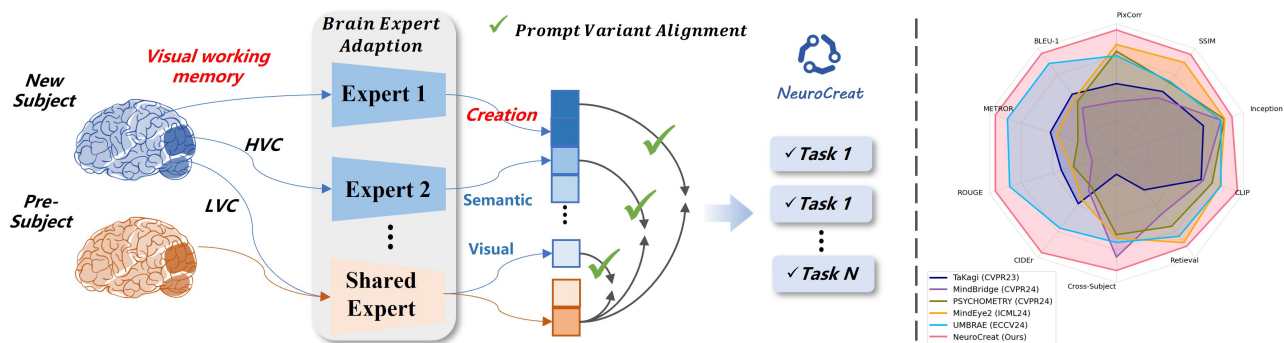


Figure 1. NeuroCreat can combine brain multi-features and LLM to achieve accurate semantic extraction with brain expert adaptation as well as prompt variant alignment, and to extract creative information from visual working memory.

Abstract

Decoding visual information from fMRI signals is an important pathway to understand how the brain represents the world, and is a cutting-edge field of artificial general intelligence. Decoding fMRI should not be limited to reconstructing visual stimuli, but also further transforming them into descriptions, creating actions, and even generating unseen content. We purposefully propose a novel and efficient brain multimodal architecture, **NeuroCreat**, which combines the powerful visual and textual abilities of LLM to capture fine-grained semantic information from fMRI, transformed it into embodied implementation of different neural representations. Specifically, we innovatively designed a **brain expert adaption** (BEA) module, effectively capturing commonalities and individual differences among subjects through the collaborative learning of shared/routed experts. Inspired by human visual working memory, we extracted “**creation**” information from higher visual cortex for idea generation. We further constructed a **prompt variant alignment** module, seamlessly integrates fMRI-visual-semantic-creation into LLM to achieve flexible incorporation of different semantics in the decoding of neural representations.

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Experiments on different fMRI datasets show that **NeuroCreat** achieves SOTA performance on multiple brain decoding tasks. More importantly, we have innovatively achieved few-shot brain video creation, which opens up a new direction for demonstrating the brain’s ‘imaginative’ ability.

1. Introduction

“**The brain’s lens not only sees but creates realities.**” The human brain is a complex and marvellous organism from which we have made advances in artificial intelligence by imitating only some simple mechanisms [47]. The most direct way to understand how the brain works is “seeing” the information from neural signals, such as fMRI [26], which encodes the world we perceive every day. Thus the brain becomes an experienced creator as we think about and witness the world [55]. The challenge of mimicking the complexity and variety of these tasks excites us [31]. We hope that the visual-semantic decoding model we have constructed will help more people in the future to demonstrate their creativity and make progress in practical application areas such as brain-computer interfaces (BCI) [9].

Humans possess a unique ability to extract semantic concepts from visual information, and the learning of diverse

concept representations has further propelled the development of human imagination. In recent years, with advancements in generative artificial intelligence and large language models, decoding visual information and semantic content from human brain signals has made many progress [39, 49, 53]. However, a key issue remains: how can we capture individual differences in the decoding process? For instance, when different people observe an image of a man surfing, some may associate it with sustained joy, while others may envision him falling into the sea. This divergence exemplifies the richness of human creativity [6, 27, 28]. We aim to integrate such diversity into our models, transforming individualized creativity into multimodal task execution capabilities, extending from the cognition of a single visual concept to multimodal decoding and creative extrapolation.

The brain’s creativity stems from the transmission and integration of information, we wish to obtain information about behavioural understanding and verbal interaction of scenes from fMRI and use fine-grained visual conceptual representations to enable flexible application of brain decoding. Revisiting the existing studies [52], we believe that it is more consistent with the path of cognitive neuroscience to take ‘visual working memory’ as the starting point of creation, and many studies have shown that visual working memory [15, 32, 33, 42] contains human predictions of short-term visual concepts. We therefore propose the “creation” task, an extension of mental creation based on fMRI signals, which aims to decode human intentional tendencies towards objects from VWM, thereby guiding the content generation of “creation”.

We propose *NeuroCreat* as shown in Fig. 1, a new framework dedicated to the realisation of creative multimodal brain decoding tasks. *NeuroCreat* consists of three main components, firstly training the brain adapter that implements shared encoding across multi-subject, **Brain Expert Adaption** (BEA) was used to achieve consistency of voxel information cross-subject, representation of the same pattern of neural representations and different individuals is achieved through shared / routed expertise. Secondly, we proposed **Prompt Variant Alignment** module, modelled the cognitive mechanism of the human brain and constructed a decoding framework by visual structural information and fine-grained semantic feature, and represent “creation” prompt from visual working memory in higher visual cortex for integration. Finally, we use visual messaging mechanisms to guide multimodal training, using LLM to integrate fMRI with visual-semantic features to achieve accurate descriptions of subjects’ visual concepts as well as creation guidance for different subjects. fMRI was mapped to the latent embedding space, used Latent Diffusion Model (LDM) [36], which was combined with the visual semantic of different subjects to achieve high-precision downstream visual tasks and creation generation. *NeuroCreat* accurately

captures and extends visual concepts in the brain, demonstrating the extended potential of multimodal decoding in the brain.

In summary, the contributions of our work are threefold:

- We propose *NeuroCreat* to effectively integrate LLM with brain decoding, a multiplexed neural decoding model that achieves unity between LLM and the multimodal visual decoding of the human brain. Excitingly, we realized a creative extension based on the visual working memory of the human brain.
- We innovatively propose the **Brain Expert Adaption** (BEA) module, it abstracts visual learning into different representations of neural modalities, enabling the sharing of visual language and specific neural representation learning cross-subject. And we designed **Prompt Variant Alignment** to realize fine-grained alignment of different layers of features, which effectively fits the human cognitive mechanism.
- We conducted experiments on different dataset [1, 18], *NeuroCreat* achieves quantitative / qualitative SOTA performance in multiple downstream tasks and innovatively implements creation generation tasks.

2. Related Work

Brain Decoding. Decoding human visual information and cognitive concepts from brain signals is a challenging task. *MinD-Vis* [8] used a combination of MAE and diffusion models achieved high-precision semantic extraction. *Mindeye* [38] demonstrated that fine-grained voxel enhancement and deep MLP could achieve high-level reconstruction. There are also some researchers trying to introduce caption of images into visual decoding [12, 27]. As the research progresses, cross-subject brain decoding became a more worthy research direction [20]. The core of the problem lies in establishing a shared representation space to efficiently align the functions of different subjects [46]. *Mindeye2* [39] utilises a small portion of fMRI for training and uses scarce data fine-tuning to achieve it. *MindBridge* [49] proposed a biologically-inspired aggregation function and a cyclic fMRI reconstruction to achieve topic-invariant representation learning. *MindArt* [7] enables visual reconstruction of different styles through graph-matched neural representations. However, we need to point out that existing cross-subject decoding methods are in latent space for the alignment of feature embeddings and mostly use a single “expert” model, which makes it difficult to learn the neural representation patterns of subjects between changing datasets. In this work, we use *brain expert adaption* to achieve efficient explicit feature variable learning by introducing MOE at each layer of the encoder.

Brain-based Multimodal Tasks. With the development of MLMM, multimodal information is incorporated in various research areas. In the field of brain decoding, in ad-

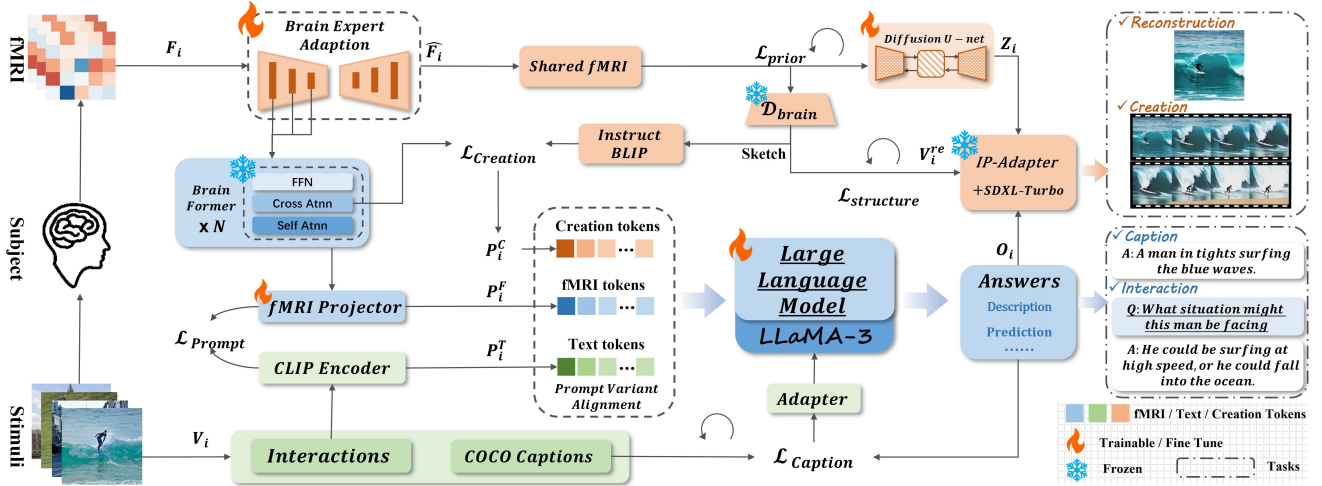


Figure 2. Overview of NeuroCreat. Brain expert adaption (BEA) achieves the preservation of individual neural representations and effective learning of shared representations. LLM is introduced to extract high-level semantic embedding features for downstream tasks. Multi-branch decoding models enable interesting tasks such as creative generation of fMRI.

dition to the traditional visual generation as the base task modality, more downstream tasks have been continuously explored. oneLLM [16] tries to introduce fMRI as a brand new modality in MLMM research. BrainCap [12] attempts to use LLM to construct visually grounded fMRI-based caption generation. UMBRAE [52] challenges to decode the grounding information of visual content from fMRI signals. MindArt [7] introduces the task of style migration through neural representations of graph matching. We have to think about what kind of tasks for multimodal interaction can really help humans manifest their intentions. In this work, we take multimodal tasks one step further, and in addition to optimally completing reconstruction and description tasks based on *prompt variant alignment*, we attempt to introduce the modal task of “*creation*” based on high-level visual concepts, for potential application extensions.

3. Methodology

The NeuroCreat framework is shown in Fig. 2. It contains three stages: Brain expert adaption (BEA) module for shared neural embedding learning, the semantic concept representation and fine-grain prompt variant alignment for LLM, the multi-modal (including “Creation”) decoding with multi-branch. We will provide more details below.

3.1. Shared Embedding Learning for BEA

Here we designed a brain expert adaption (BEA) module as cross-subject adaptor to achieve visual-semantic feature consistent learning. This efficiently embeds voxel information from different subjects into shared expert mappings and provides a robust method for fMRI data scarcity.

We defined the combination of fMRI and stimulus image (*fMRI, Image*) as $\mathbf{S} = (\mathbf{F}_i, \mathbf{V}_i)_{i=1}^n$, $\mathbf{F}_i \in \mathbb{R}^{1 \times N}$, $\mathbf{V}_i \in \mathbb{R}^{H \times W \times 3}$. As shown in Fig. 3, We redesigned the feature

extractor to first encode the images using different unimodal encoder, and then jointly characterise the features to obtain a visual continuous embedding by a causal transformer. We use shared expert and routed expert as processors specific to each subject. Shared expert continuously captures commonalities in cross-subject learning, and routed expert is used to learn each subject-specific parameter to represent individual representations of neural differences. Thus, the learning of underlying neural properties for fMRI and images constitutes the complete BEA module. When the new subject data input the adaptor to do cross-subject processing, We can obtain the adapted fMRI representation:

$$\hat{\mathbf{F}}_i = \text{Expert}^S(\mathbf{F}_i) + \sum_{i=1}^{N-1} (\text{Expert}^R(\mathbf{F}_i)), \quad (1)$$

where $\text{Expert}^S(\cdot)$ is a shared expert model to learn a uniform neural representation and $\text{Expert}^R(\cdot)$ is a routed expert model to characterise neural differences across subjects, BEA set expert architecture as previous work. We performed gradient updates for parameters of BEA weight matrix $\mathbf{G} \in \mathbb{R}^{d_f \times d_v}$, where d_f and d_v denotes the fMRI/visual dimension, and we can use the new subject $\mathbf{F}_i$ to constrained the gradient updates:

$$\hat{\mathbf{G}} = \mathbf{G} + \mathbf{H}(\mathbf{F}_i), \quad (2)$$

$\mathbf{H}(\cdot)$ is the learnable matrix for BEA, it is made up of $\mathbf{Z}_f^{\text{share}}$ and $\mathbf{Z}_f^{\text{route}}$ parts. At begin, we initialised the parameters of matrix $\mathbf{G}$ by initiating variables to pre-train values for better results for each adaption.

Thus, we can obtain the fMRI features $\mathbf{Z}_f^i = E_f(\hat{\mathbf{F}}_i)$ after BEA adaption and the fused multimodal continuous embedding features $\mathbf{Z}_v^i = E_v(\mathbf{V}_i)$ of images. We input $\mathbf{Z}_v^i$ tokens to brain embedder layer #1 for dimensionality unification so that comparative learning with expert tokens

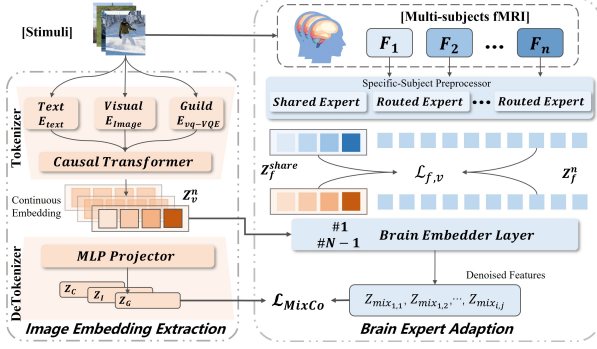


Figure 3. brain expert adaption (BEA) Module. Enabling learning of shared and unique neural representations.

can be done. When data from new subject input into BEA, we used the embedding relationships to adapt all mapping layers across experts accordingly, using the loss from CLIP-Expert and Shared-Routed scenarios as $\mathcal{L}_{f,v}$ to update features Z_f^{share} tokens enable shared expert learning:

$$\mathcal{L}_{f,v} = -\frac{1}{N} \sum_{i,j=0}^{N-1} \left[\frac{\exp(\frac{Z_f^i \cdot Z_f^j}{\tau})}{\sum_{k=0}^{N-1} \exp(\frac{Z_f^i \cdot Z_f^k}{\tau})} \cdot \log\left(\frac{\exp(\frac{Z_f^i \cdot Z_f^j}{\tau})}{\sum_{k=0}^{N-1} \exp(\frac{Z_f^i \cdot Z_f^k}{\tau})}\right) \right], \quad (3)$$

where τ is the temperature hyperparameter.

Then we look back at the complete process of encoding the image to get the features, we encode the visual stimuli as textual and visual features of CLIP and use VQ-VAE for guide space encoding. Here we utilise Z_C^i , Z_I^i , Z_G^i to represent the extracted features. Z_C^i and Z_I^i are used to decode the high-level semantic features and the low-level structural features of images. Z_G^i used to characterise the process of visual guidance, which is used to model the process of top-down control of human vision. We use $Z_F^i = Z_C^i + Z_I^i + Z_G^i$ to represent the overall fused features of the image. For the output features of the brain embedding layer $\#N-1$, we use the effective voxel mixing paradigm MixCo to represent it, with the weight of shared Expert^s($\cdot$) as factor η :

$$Z_{mix_{i,j}} = MLP(\eta_i \cdot \hat{F}_i + (1 - \eta_i) \cdot \hat{F}_j), \quad (4)$$

where j represents the arbitrary blend index of the batch, and $Z_{mix_{i,j}}$ is L2-normalised. Then we constructed loss $\mathcal{L}_{Align} = \mathcal{L}_{Ex|MixCo}$ with the mix-up Contrast loss and expert weights as the cross-modal alignment loss:

$$\mathcal{L}_{Align} = -\sum_{i=0}^{N-1} \left[\eta_i \log f \frac{\exp(Z_{mix_{i,j}} \cdot Z_F^i / \tau)}{\sum_{m=0}^{N-1} \exp(Z_{mix_{i,j}} \cdot Z_F^m / \tau)} + (1 - \eta_i) \log f \frac{\exp(Z_{mix_{i,j}} \cdot Z_F^j / \tau)}{\sum_{m=0}^{N-1} \exp(Z_{mix_{i,j}} \cdot Z_F^m / \tau)} \right], \quad (5)$$

where τ is the temperature hyperparameter. Then we can get cross-subject adaption loss:

$$\mathcal{L}_{Cross} = \mathcal{L}_{f,v} + \mathcal{L}_{Align}. \quad (6)$$

BEA ensures shared expert achieves a representation of shared features on each subject and routed expert achieves preservation of individual differences. Voxel-level alignment is achieved and individual differences are not ignored.

3.2. Prompt Variant Alignment for LLM

Prompt information from fMRI is the core of introducing various modalities and decoding them accurately, and the mapping of existing methods ignores synthetic matching within the prompt. In NeuroCreat, we design prompt variant alignment to separate the differences of different modalities, which can effectively reduce the problems of low fMRI resolution and excessive modal differences, and help to align multiple modalities.

We designed BrainFormer using an MLP projector and N transformer modules as the backbone to extract the representation of fMRI into the textual representation space as the fMRI prompt P_i^F . We designed a series of Interaction questions for the stimulus images as instructions, which are utilized to obtain high-level semantic information in fMRI, and to complete complex language tasks, we used CLIP [35] last hidden layer to encode it as the text prompt P_i^T . And we extract the middle layer information of BrainFormer and visual working memory brain region ROI for cross-attention computation. Using the obtained generated embedded features to generate $P_i^{C'}$. Meanwhile, we used InstructBLIP for basic reconstruction V_i' of images to predict possible future behaviours as the creation prompt P_i^{C*} . Then we apply alignment loss $\mathcal{L}_{Cre}$ to obtain accurate creative P_i^C . The loss function is defined as follows:

$$\mathcal{L}_{Cre} = -\frac{1}{|\tau|} \sum_{F_i, V_i} \mathcal{KL}(P_i^{C'}, P_i^{C*}), \quad (7)$$

Then we can compute the cross-entropy loss $\mathcal{L}_{Cap}$ on the generated Caption $Cap^{T'}$ and the image caption Cap^T for the variant alignment of P_i^T :

$$\mathcal{L}_{Cap} = -\frac{\sum_{i=0}^{T-1} \log P_\theta(Cap^T | t < i, N)}{\sum_{i=0}^{T-1} \|Cap^T - Cap^{T'}\|_2}, \quad (8)$$

where t represents the text, T represents the total time step, and $P_\theta(Cap^T | t < i, N)$ represents the probability of generating Cap^T at specified time step and fMRI conditions.

The prompts extracted from fMRI may be noisy and difficult to align, thus further fine-tune using textual features is required. We put the cross-modal adapted fMRI into fMRI Projector (includes 1D convolutional layer, fully-connected layer and Attentional Pooler) to generate the fMRI prompt, and then align the prompts P_i^F using the loss $\mathcal{L}_{Prompt}$:

$$\mathcal{L}_{Prompt} = -\sum_{i,j=0}^{N-1} [\log P_i^F(tok_j) + \log P_i^T(tok_j)], \quad (9)$$

where tok represents the tokens of the text/fMRI and I is

the length of the prompt. The total loss of prompt variant alignment is:

$$\mathcal{L}_{NeuroCreat} = \mathcal{L}_{Cre} + \lambda_{Cap}\mathcal{L}_{Cap} + \lambda_{Prompt}\mathcal{L}_{Prompt}, \quad (10)$$

where λ_{Cap} and λ_{Prompt} is the weighted hyperparameter.

In the inference, we use a similar effective regression form to generate answers using LLM. We can represent the process as: $User : [fMRI][Text][Creation] \rightarrow LLM : [Answer]$, where $[Text]$ and $[Answer]$ serve as placeholders for interaction input and LLM output. $[Creation]$ provides a creative behaviour for reconstructed image.

In conclusion, based on the above inputs and settings to the LLM, NeuroCreat implements downstream tasks such as generating creations, captions, detailed descriptions, and complex reasoning, which provide an adequate basis for fMRI-based vision tasks. The generic framework of NeuroCreat enables the use of different encoders and LLMs.

3.3. Multimodal Visual Task Implementation

In the visual task implementation, the process is shown in Fig. 2. From the point of view of simulating human visual cognitive process, NeuroCreat fully considers the flow of information and fits human low-level visual cognition (sketch), high-level visual cognition (feature embedding), and semantic conceptual cognition. And it creatively extracted the ability of human creation from visual working memory, generating action prediction after perceiving the scene, stretching the brain’s creativity.

Specifically, first we use the sketch V'_i reconstructed based on the pre-trained VAE as the basis of our reconstruction, which is also the basis of the creatively generated action prediction. Secondly, the embedding of our diffusion process is obtained by a streamlined Diffusion U-Net, which can better achieve zero-shot image classification. We perform cross-attention coding on the penultimate text encoding, embedding it as a time step into the LDM, as high-level visual feature Z_i . Finally, we make full use of the description information O_i generated by the LLM, which contains information such as caption, prediction, which we use as the semantic a priori knowledge for reconstruction.

The main component of visual generation by fMRI is mapping fMRI $\hat{F}_i$ to image V_{re} , so we can define the task of image reconstruction as: $D_{Brain}(\hat{F}_i) \rightarrow V_i^{re} = Diffusion(V'_i + Z_i + O_i)$.

We can express the reconstruction loss as the sum of the MSE loss and the structural loss between V_i^{re} and V_i :

$$\mathcal{L}_{reconstruct} = \alpha\mathcal{L}_{Prior} + \beta\mathcal{L}_{Structure}. \quad (11)$$

Then we continuously update the quality of the image generation in each stage through backpropagation until convergence, to achieve a well-controlled D_{Brain} .

For D_{Brain} here we used SDXL-Turbo [37] as the generative model, which effectively improves the speed of image

synthesis. At the same time, we use IP-Adapter, which effectively enhances the cue adaption of LLM-generated textual information during image generation.

4. Experiments

4.1. Experimental Setup

We used two publicly available fMRI visual stimulus datasets: **NSD** dataset [1] consists fMRI of 8 subjects as they viewed COCO [24] stimuli. Each subject viewed 10,000 images, 9,000 special and 1,000 shared images. We used the 4 subjects completed all scans: S1, S2, S5 and S7. The training set for each subject contains 8,859 images and 24,980 fMRI trails, the test set contains 982 images and 2,770 fMRI trails. We used ‘nsdgeneral’ to preprocess [38, 44] the voxels and used the COCO caption. **GOD** dataset [18] consists fMRI of 5 subjects as they viewed ImageNet stimuli, the training set contains 1,200 images (8 images from each of 150 different categories) and test set consists of 50 images (1 from each of 50 categories). Each subject resulting in 1,200 and 1,750 fMRI trials. We acquire caption labels from the GOD-Cap dataset [25], offering two descriptions for each image.

Architecture. We used the pre-trained CLIP ViT-L/14 & bigG-14 [35] as visual encoder. LLM model chose LLaMA-3-8B [29] with input dimension of 4,096, it was fine-tuned using the LLaMa Adapter [54]. In the encoded output of the penultimate layer of textual attention, we used 256 patch tokens associated with fine-tuned features.

Implementation. NeuroCreat is trained on 4 NVIDIA RTX 4090. The global batch size is 256 for 500 epochs of training, for learning rate tuning we use one-cycle [41] with initial learning rate of 2×10^{-4} . For cross-subject training, with an alignment dimension of 12,490 per subject, we set 64 of the 256 samples come from the shared expert, the rest for adaption with 200 epochs. For fine-tune, we used 2 RTX 4090 with batch size of 32. For visual task, we used 50 diffusion steps with intensity of 0.75.

Metrics. In visual task, we used widely accepted metrics, low-level: PixCorr, SSIM [50], AlexNet(2 & 5) [21], high-level: Inception [43], CLIP [35], EffNet-B [45], and SwAV [4]. In captioning and interaction task, the metrics we adopted are: BLEU- k [30], METEOR [3], ROUGE-L [23], CIDEr [48], SPICE [2], CLIP-S [17]. In creation task, we used the latest aigcbench [10] (e.g. Flow-Square-Mean, GenVideo-RefVideo Clip) to evaluate the quality of creation generation, and we demonstrated fMRI-based creations using actions in coherent frames.

4.2. Comparison with State-of-the-Art Methods

Visual Reconstruction and Retrieval. NeuroCreat enables cross-subject decoding using a single model, achieving the goal of few-shot learning using less fMRI data. Quantitative

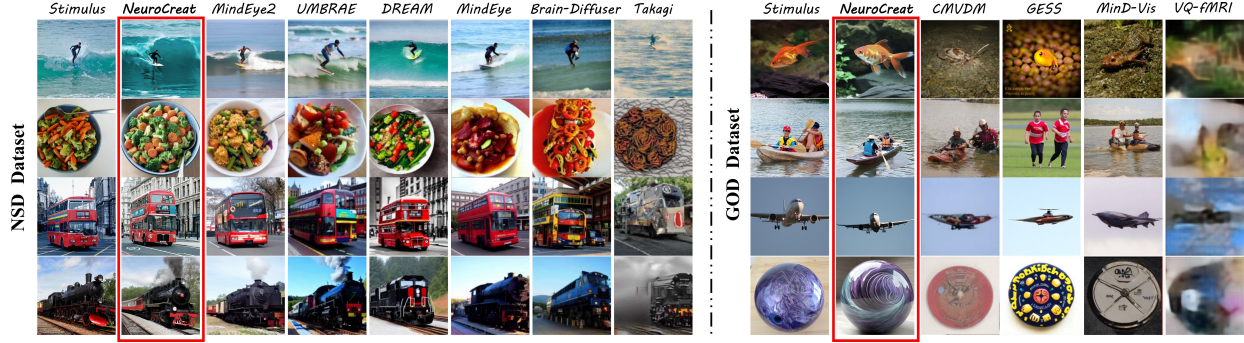


Figure 4. Visual reconstruction on NSD and GOD dataset. NeuroCreat decodes the most accurate images with semantic details.

Dataset	Method	Low-Level				High-Level				Retrieval	
		PixCorr $\uparrow$	SSIM $\uparrow$	Alex(2) $\uparrow$	Alex(5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	EffNet-B $\downarrow$	SwAV $\downarrow$	fMRI $\uparrow$	Image $\uparrow$
NSD	Takagi. [44] [CVPR'23]	-	-	83.00%	83.00%	76.00%	77.00%	-	-	-	-
	MindEye [38] [NeurIPS'23]	.309	.323	94.70%	97.80%	93.80%	93.30%	.648	.367	93.60%	90.1%
	DREAM [51] [WACV'24]	.274	.328	93.90%	96.70%	93.40%	94.10%	.645	.418	-	-
	+BEA(%)	.299(<u>9.1</u>)	.345(<u>15.1</u>)	96.70%	98.10%	95.20%	95.00%	.625(<u>13.1</u>)	.363(<u>113.2</u>)	-	-
	MindBridge [49] [CVPR'24]	.151	.263	87.70%	95.50%	92.40%	94.70%	.712	.418	-	-
	PSYCHOMETRY [34] [CVPR'24]	.297	.340	96.40%	98.60%	95.80%	96.80%	.628	.345	-	-
	MindEye2 [39] [ICML'24]	.322	.431	96.10%	98.60%	95.40%	93.00%	.619	.344	98.30%	98.80%
	UMBRAE [52] [ECCV'24]	.283	.341	95.50%	97.00%	91.70%	93.50%	.700	.393	94.20%	91.30%
	Neuro-V [40] [NIPS'24]	.265	.357	93.10%	97.10%	96.80%	97.50%	.633	.321	-	-
	NeuroCreat (Subject-1)	.318	.411	96.10%	97.70%	<u>96.10%</u>	95.40%	.622	.343	97.10%	95.90%
GOD	NeuroCreat	.339	.448	97.10%	98.90%	<u>96.00%</u>	97.80%	.602	<u>.331</u>	98.80%	98.90%
	MinD-Vis [8] [CVPR'23]	.120	.389	82.80%	93.80%	81.10%	82.60%	.836	.491	-	-
	VQ-fMRI [5] [ICML'23]	.180	.423	87.20%	92.70%	78.70%	74.80%	.838	.498	-	-
	GESS [11] [NeurIPS'23]	.131	.341	81.60%	91.40%	79.80%	72.10%	.841	.493	78.40%	<u>81.00%</u>
	CMVDM [53] [AAAI'24]	.279	.455	88.30%	94.00%	81.70%	82.10%	.810	.485	<u>79.10%</u>	75.40%
	NeuroCreat	.323	.501	91.70%	95.70%	89.80%	87.30%	.787	.467	86.70%	90.20%

Table 1. Quantitative evaluation on visual reconstruction. Results are calculated from the average of subjects, with **bold** indicating the best results, underlining the second best results, **+BEA(%)** represents frozen the model and adding the BEA module.

analyses are shown in Tab. 1, we compare NeuroCreat with previous reconstruction methods. We report the average results of 4 subjects and the results trained only on subject-1 (S1). The results show that NeuroCreat achieves SOTA across nearly all metrics (e.g. SSIM 33.9%, CLIP 97.8%), demonstrating its ability to accurately capture semantic information and visual structure of stimuli from fMRI. We also note that NeuroCreat learned cross-subject performed better than only on S1, suggesting a role for BEA shared expert. Training only on S1 still achieves good results suggesting the routed expert learns the unique neural representations. In the retrieval results, NeuroCreat achieved the optimal accuracy of 98.4% and 98.8%. The retrieval task helps to evaluate the fine-grained information of stimuli contained in fMRI embeddings. The results demonstrate NeuroCreat achieves effective capture of fine-grained features.

Visualisation results of reconstruction are shown in Fig. 4. Our performance is impressive, achieving highly consistent results with the original images, both in terms of capturing detailed features: structure, location, colour, as well as semantic features: object, content. The high-precision reconstruction of visual features can show that NeuroCreat fine-grained control reconstruction process using multi-level visual feature guidance is very effective.

	MSE $\downarrow$	SSIM $\uparrow$	FVD $\downarrow$	FSM $\uparrow$	G-Ref-Clip $\uparrow$	GV-Clip $\uparrow$	DOVER $\uparrow$
UMBRAE+AIGC	23.66	15.25	58.77	1.44	.647	.788	.664
Ours(NSD S-1)	18.15	18.34	51.53	1.73	.778	.947	.798
Ours(NSD)	17.29	19.31	49.08	1.82	.819	.997	.840
Ours(GOD)	21.96	24.52	62.33	2.31	1.040	1.266	1.067

Table 2. Quantitative evaluation of creation generated videos.

Visual Creation. We visualise the results of the visual creation on the Fig. 5 left, NeuroCreat performed the video generation based on the predicted next movements of the images. To visually demonstrate NeuroCreat’s decoding creativity, we intercepted the frames of the video where the object is moving for demonstration and showed the predicted next action content. From the results, we can clearly see that the creative video generation matches the semantic content of the predicted actions, and the generated video frames are clear and smooth. In Tab. 2 we quantitatively analyze the Creation results, and since there is no related work based on NSD, we use the visual reconstruction results from UMBRAE and the AIGC model [19] cascade for comparison. We also show the performance on Subject-1, NSD and GOD datasets, and we can see that NeuroCreat achieves excellent metrics, much higher than the simple **“Reconstruction + Generation”** approach. NeuroCreat not only achieves the optimal image reconstruction and semantic information retention, but also effectively applies the

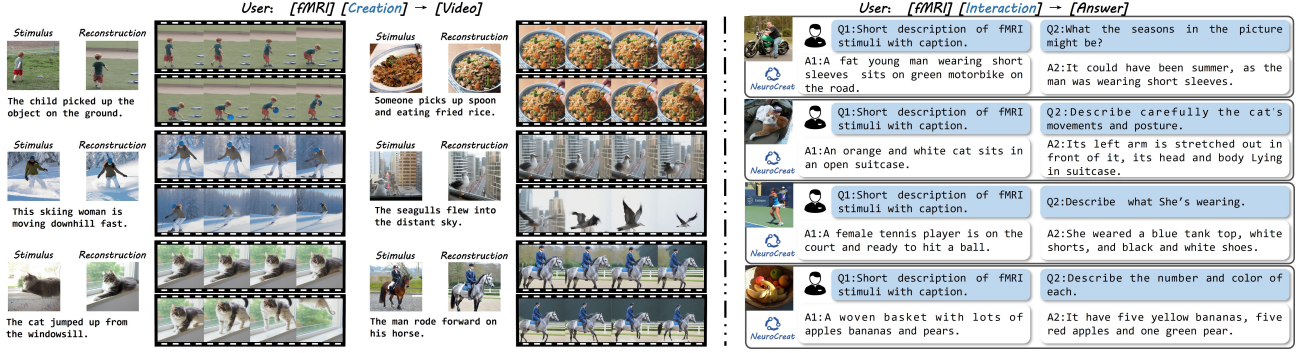


Figure 5. Examples of textual task and creation task results. NeuroCreat can implement downstream tasks such as caption generation and linguistic interaction for fMRI. It can also predict future actions and generate creation videos based on fMRI reconstruction.

	Method	B@1↑	M↑	R-L↑	C↑	S↑	CLIP↑
Caption	BLIP-2 [22] [ICML'23]	73.79	33.90	57.34	98.23	28.26	70.13
	BrainCap [12] [arxiv'23]	55.96	16.68	40.69	41.30	9.06	64.31
	MindGPT [6] [arxiv'23]	37.90	12.80	35.90	70.80	10.30	-
	MindArt [7] [CVPR'24]	39.10	15.60	39.70	-	11.00	-
	OneLLM [16] [CVPR'24]	47.04	13.55	35.05	22.99	6.26	54.80
	+BEA(%)	52.16(↑10.9)	16.01(↑18.2)	37.20(↑6.1)	30.78	9.11	60.93(↑11.2)
	UMBRAE [52] [ECCV'24]	59.44	19.45	43.71	61.06	12.79	67.78
	Neuro-V [40] [NIPS'24]	57.19	18.60	36.67	49.51	12.39	65.49
	NeuroCreat (S-1)	57.23	19.73	38.69	65.12	11.32	65.44
	NeuroCreat	60.01	20.22	45.11	69.93	15.90	69.31
Inte	NeuroCreat (S-1)	60.19	31.51	44.90	88.17	30.24	73.07
	NeuroCreat	66.24	38.60	49.38	96.21	32.39	79.91

Table 3. Quantitative analysis of brain captioning and interactions.

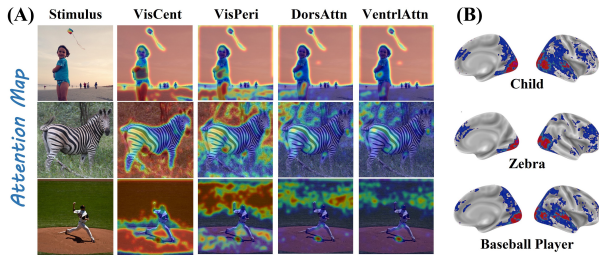


Figure 6. (A), Projections of attention represented by different brain regions (VisCent: Visual Central Net, VisPeri: Visual Peripheral Net, DorsAttn: Dorsal Attention Net, VentrlAttn: Ventral Attention Net). (B), Representation of NeuroCreat hidden layer projections on brain with significant activation in visual cortex.

conceptual knowledge learnt from fMRI to build creative video content that can be interacted with by human. NeuroCreat extends the tasks and functions of brain decoding.

Captioning and Interaction. In Tab. 3, We report results comparing NeuroCreat with various methods on captioning and interaction task. NeuroCreat achieves SOTA in all semantic metrics, and is especially able to ensure the excellence of high-level metrics, which provides more flexibility for downstream tasks of complex linguistic environment, such as action creation, shows the excellent semantic capture abilities. In addition, we can see that cross-subject trained NeuroCreat can get better results, which suggests that the shared expert’s semantic space is more conducive to high-precision information extraction. Meanwhile, we

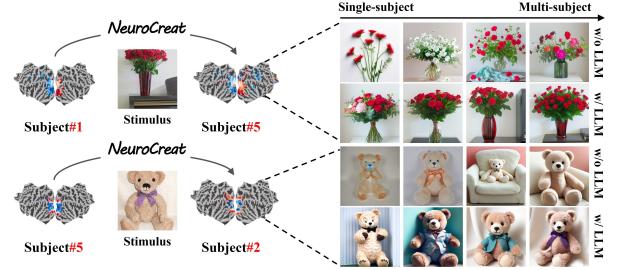


Figure 7. Visualisation of cross-subject fMRI adaption and visual reconstruction in different conditions using BEA.

Model	PixCorr↑	SSIM↑	Alex(2)↑	Inception↑	CLIP↑	EffNet-B↓	SwAV↓
w/o P_i^T	.311	.422	94.10%	81.20%	82.50%	.694	.378
w/o P_i^F	.293	.401	85.30%	94.10%	95.40%	.632	.355
w/o P_i^C	.327	.435	96.20%	95.40%	96.50%	.617	.339
Ours	.339	.448	97.10%	96.00%	97.80%	.602	.331

Table 4. Ablation results for prompt variant alignment.

use BLIP-2 [22] to generate captions as an upper bound.

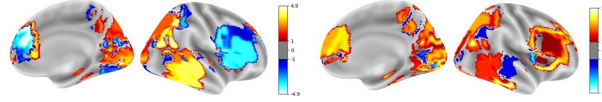
We visualise some of the caption and interaction results in Fig. 5 right. It can be observed that the caption and answers to questions generated by NeuroCreat matches well with the stimuli, and the flow of the utterance is smooth.

4.3. Neural Representation and Visualisation

In Fig. 6 left shows the attention area projection of the fMRI from different brain regions network in the task onto stimuli. It is easy to see that the VisCent/VisPeri region is more significant for attentional representations of semantic object in stimuli. On Fig. 6 right we show the results of projecting the potential attention of the NeuroCreat decoded hidden layer onto the cerebral cortex, and it can be seen that the activation of the visual cortex is significant, which effectively demonstrates that the NeuroCreat is reliably biologically interpretable and can learn common semantic concepts and different neural pattern of different subjects.

4.4. Ablation & Discussion

Impact of BEA. We show the experimental results of “Dream+BEA” and “OneLLM+BEA” in one row in Tab. 1



The child picked up objects on the ground. The cat jumped up from the windowsill.
Figure 8. Creation token brain projection.

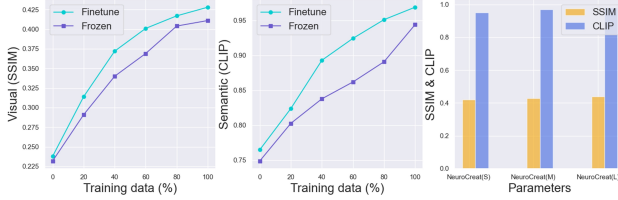


Figure 9. Task performance for different training data and parameters models (S:8 layers, M:16 layers, L:32 layers).

Method	MSE↓	SSIM↑	FVD↓	FSM↑	G-Ref-Clip↑	GV-Clip↑	DOVER↑
FFA	12.57	13.54	57.81	1.36	.682	.879	.731
LVC	14.78	15.62	55.18	1.51	.747	.928	.768
HVC	17.119	19.06	50.11	1.67	.808	.977	.827

Table 5. Results of creation using different visual brain regions.

and Tab. 3. It can be seen that after adding **BEA** to the original model, both visual and semantic metrics are significantly improved. It effectively illustrates that the cross-subject learning ability of BEA improves the extraction and learning of the model for latent features of fMRI.

In Fig. 7 left, we show the fMRI adaption results by BEA, we visualised brain activity using pycortex [14]. It can be seen that NeuroCreat achieves the maintenance of activated regions and the representation of individual voxel activation differences. This balance between commonality and specificity has helped NeuroCreat to effectively learn and characterise neural voxel activity across subjects. In Fig. 7 right, we show the results of visual reconstruction of adapted fMRI. BEA can keep the semantics unchanged during adaption. And the images reconstructed by models pretrained with single to multi-subject, shown that cross-subject co-training is also a meaningful design.

Impact of Prompt variant alignment. In Tab. 4 we show the ablation results for prompt variant alignment. We can see that P_i^T has an important role in learning advanced semantic features, and the elimination of P_i^F affects the representation of visual details. The P_i^C further complements the visual-semantic representation. Each of the Prompts exerts fine-grained control over the downstream task, and the contrast of Prompt domains with different attributes effectively improves the brain modality distinction.

Creation Analysis. In Tab. 5, we performed quantitative experiments on creations using different visual brain regions (FFA, LVC and HVC). It can be seen that different brain regions contribute differently to creations and that HVC does lead to better generative performance. In Fig. 8, we computed cross-attention by taking generated creation

Model	PixCorr↑	SSIM↑	Alex(2)↑	Inception↑	CLIP↑	EffNet-B↓	SwAV↓
w/o $\mathcal{L}_{Stru}$	.192	.276	84.70%	94.50%	94.10%	.628	.357
w/o $\mathcal{L}_{prior}$	.182	.253	85.10%	94.10%	93.80%	.629	.361
w/o $\mathcal{L}_{Cross}$	.308	.407	95.30%	94.40%	95.40%	.621	.349
Ours	.339	.448	97.10%	96.00%	97.80%	.602	.331

Table 6. Results for visual reconstruction with different loss.

$\mathcal{L}_{Cap}$	$\mathcal{L}_{Pro}$	$\mathcal{L}_{Cross}$	B@1↑	M↑	R↑	C↑	S↑	CLIP↑
✓			51.33	17.21	34.98	54.30	9.02	62.37
✓	✓		56.90	18.79	42.43	62.25	12.55	65.12
✓	✓	✓	60.01	20.22	45.11	69.93	15.90	69.31

Table 7. Results for caption with different loss.

tokens and stimuli, projected attention map onto the cerebral cortex using FreeSurfer [13]. High performance and activation of the higher visual cortex shows the consistency of the created content with the brain’s visual working memory theory, validating the interpretability of idea generation.

Ablation of Loss Functions. We show the results of ablating different losses in Tab. 6 and Tab. 7. It can be seen that adding $\mathcal{L}_{structure}$ and $\mathcal{L}_{prior}$ helps the model to learn the the image detailed features in targeted ways, adding $\mathcal{L}_{Cap}$ and $\mathcal{L}_{Prompt}$ helps to enhance the representation of semantic information and the generation of semantic content. And adding $\mathcal{L}_{Cross}$ helps to significantly improve the learning of fine-grained representations of semantic and visual information to achieve small-parameter fine-tuning.

Data numbers and Model size. We show in Fig. 9 the effect of using encoders trained with different amounts of data and different parameters on metrics. NeuroCreat achieves comparable performance using only a little data, and the performance gradually improves as data amount increases. And with only a small number of parameters encoder, NeuroCreat is still able to achieve excellent performance. This effectively demonstrates that BEA allows subjects to be adapted using minimal data, with extracts the maximum information and variance from the neural representations.

5. Conclusion

In this study, we propose a novel brain signal decoding method **NeuroCreat**, which aims to extend fMRI-based multimodal vision tasks. Inspired by the distributed processing system of the human brain, through brain expert adaption, we utilize MOE to make each expert a valid subject brain learner, providing a robust method for unified brain decoding. A key innovation of the framework is prompt variant alignment, which establishes brain-modality mappings at a fine-grained level, incorporating visual working memory embedded in higher visual cortex and mapping it accurately to “Creation” information, NeuroCreat enhances the modeling of neural features. The deployment of NeuroCreat represents a significant brain decoding advance, providing a new perspective on obtaining meaningful unbiased interpretations from the brain.

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